2025-08-04 - Analyze the data and perform interval testing for sampling distributions.

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⇒ Poisson Distribution :

Suppose you are counting the number of occurrences of an event in a given unit of time (distance, area, volume). we use poisson distribution.

Eg : No. of car accidents in a day.

No. of tulips in a square metre plot of land.

⇒ Events are occuring randomly & independently.

⇒ Then X, the number of events in a fixed unit of time has a poisson distribution.

⇒ The Poisson Disⁿ

$$\boxed{P(X = x) = \frac{\lambda^{x} \cdot e^{-\lambda}}{x!}}$$

for $x = 0, 1, 2 \ldots \ldots n$

$$\lambda = \mu \;\; \& \;\; \sigma^2 = \lambda \;\; \text{only} \;\; \sigma = \sqrt{\lambda}$$

$\Rightarrow$ One nanogram of Plutonium-239 will have an average of 2.3 radioactive decays per second & number of decays will follow a Poisson dist^n. What is the probability that in a 2 second period there are exactly 3 radio-active decays?

$\Rightarrow$ Let X represent the no. of decays in a 2 second period.

$$\lambda = 2.3 \times 2 = \boxed{4.6}$$

$$P(X=3) = \frac{\lambda^x \cdot e^{-\lambda}}{x!} = \frac{4.6^3 \cdot e^{-4.6}}{3!} = \underline{\underline{0.163}}$$

$0.163 \approx \underline{\underline{16.3\ \%}}$